INS3441

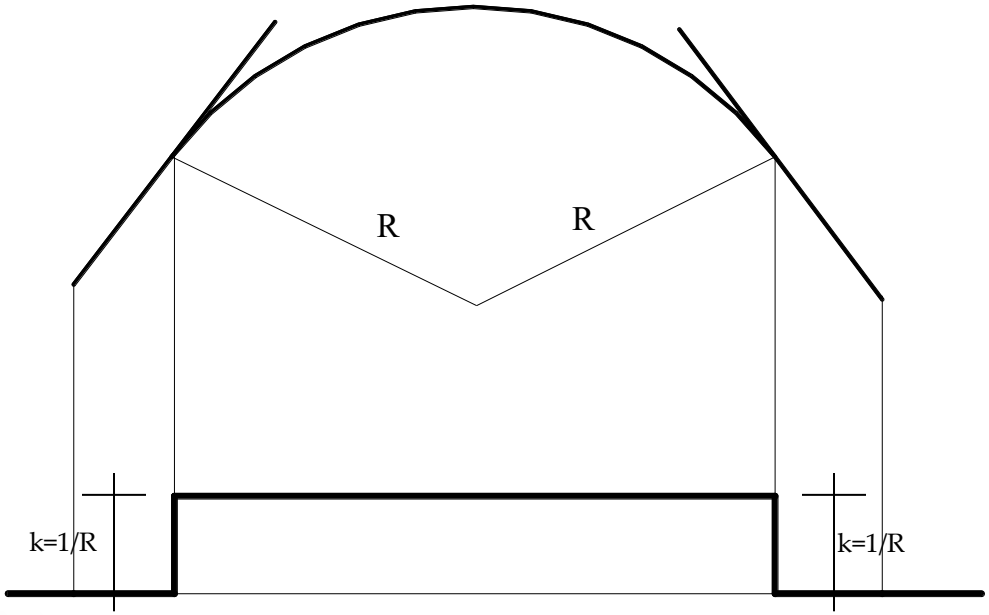
HIGHWAY ENGINEERING

Prof. Halit ÖZEN, PhD.

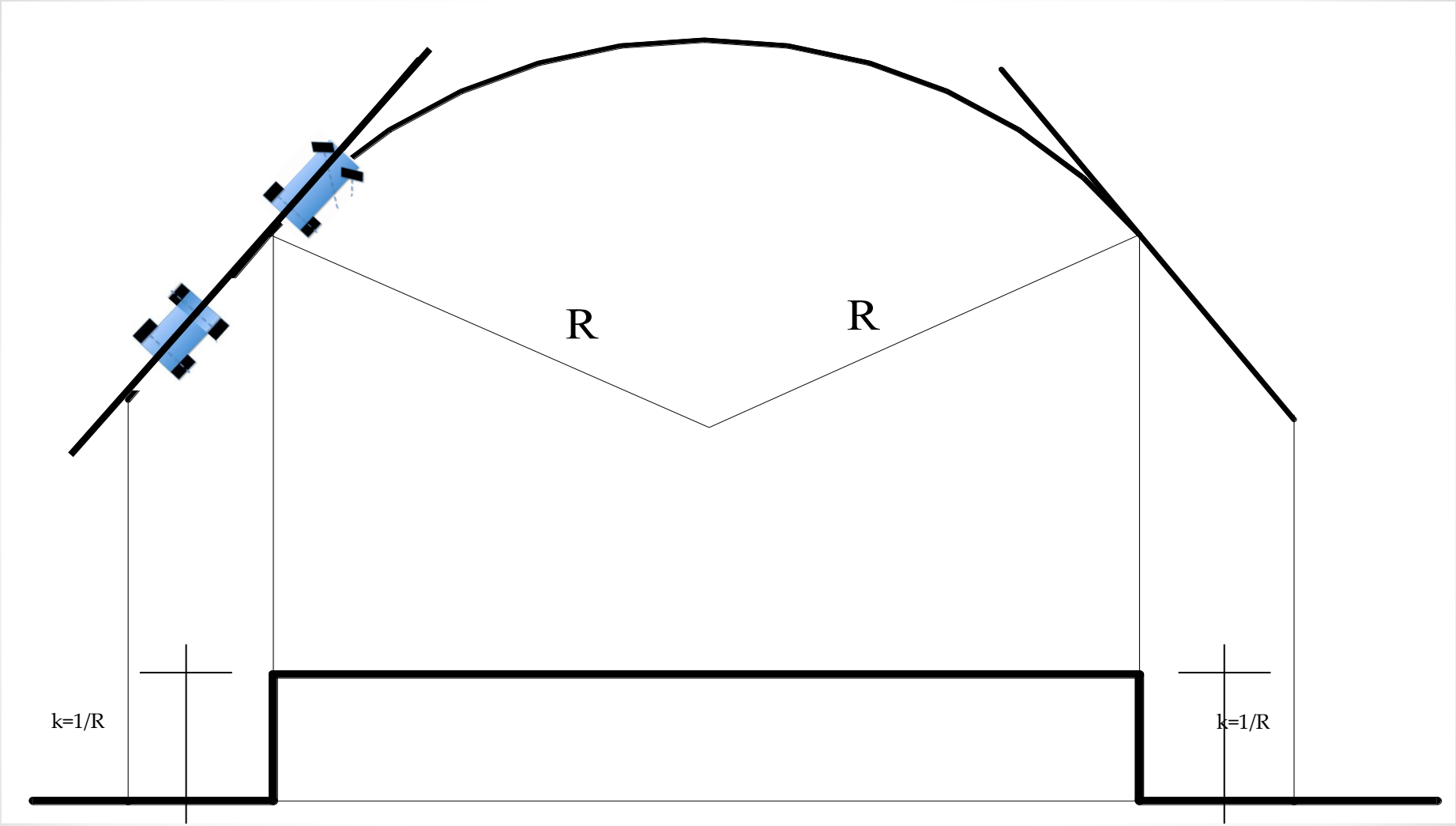
TRANSITION CURVES

Additional highway components which are placed between the allignment and the horizontal curve in order to resolve the sudden effect of the centrifugal force are named as the **transition curves**.

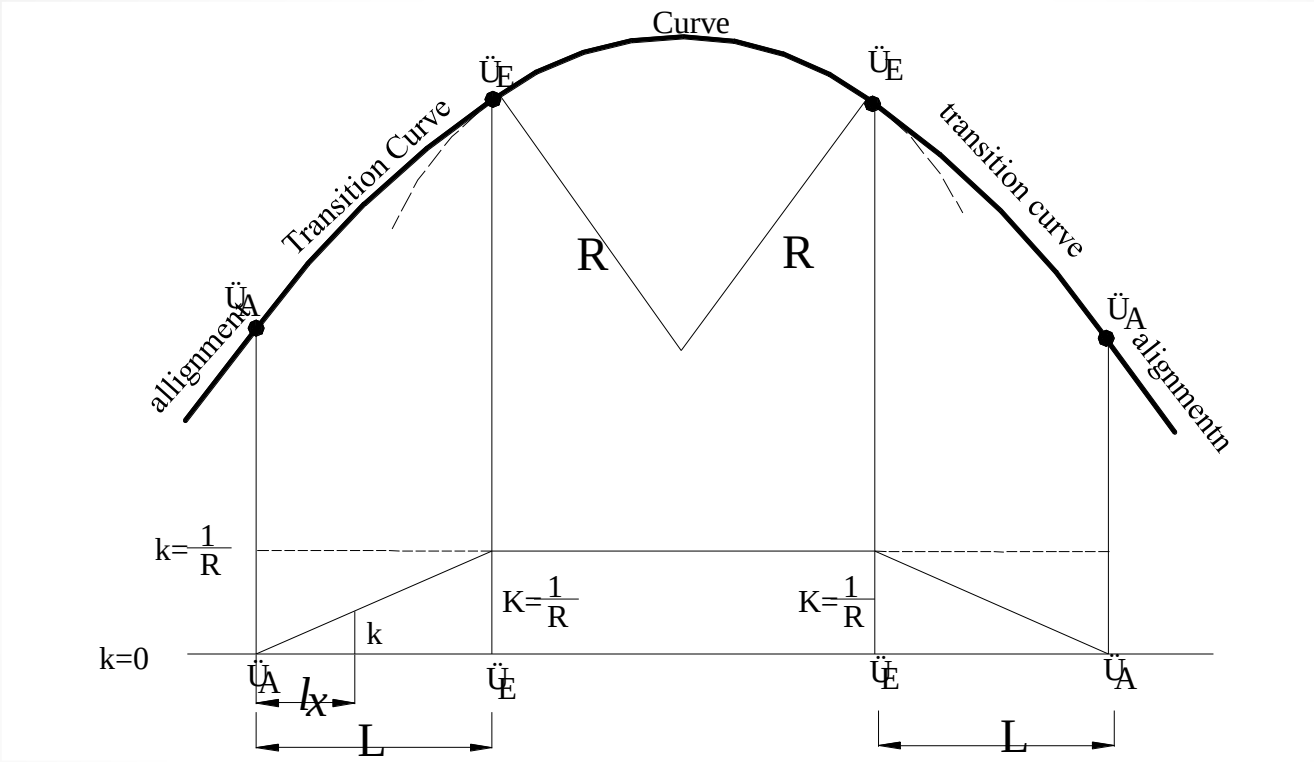
k defined as cuvature that changes in turning radii or angle.



TRANSITION CURVES



TRANSITION CURVES



$$k = \frac{K}{L} * l_x$$

$$K = \frac{1}{R}$$

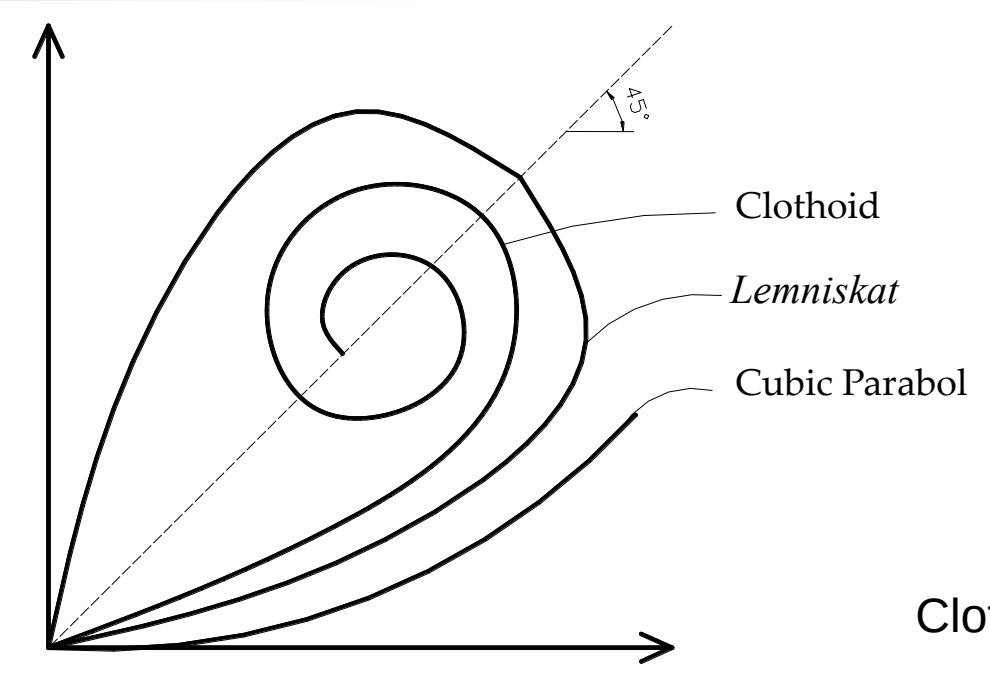
$$k = \frac{l_x}{L * R}$$

TRANSITION CURVES

The transition curve has to provide the following conditions.

1. The start of the transition curve is tangent to the allignment while its end is tangent to the horizontal curve.
2. The radius of curvature ($R_t=\infty$) is infinite at the beginning of the transition curve while it is equal to the radius of the horizontal curve ($R_t=R$) at the end. This decrease on the radius of curvature, or in other words the increase on the curvature (k) has to be linear through the curve as seen on the figure.
3. The vehicle must be able to continue its course with the constant velocity, V through the transition curve.
4. When the steering wheel of the vehicle is rotated with a constant angular velocity of ω, the highest rotation angle has to be obtained at the end of the transition curve or in other words at the beginning of the horizontal curve.

TRANSITION CURVES

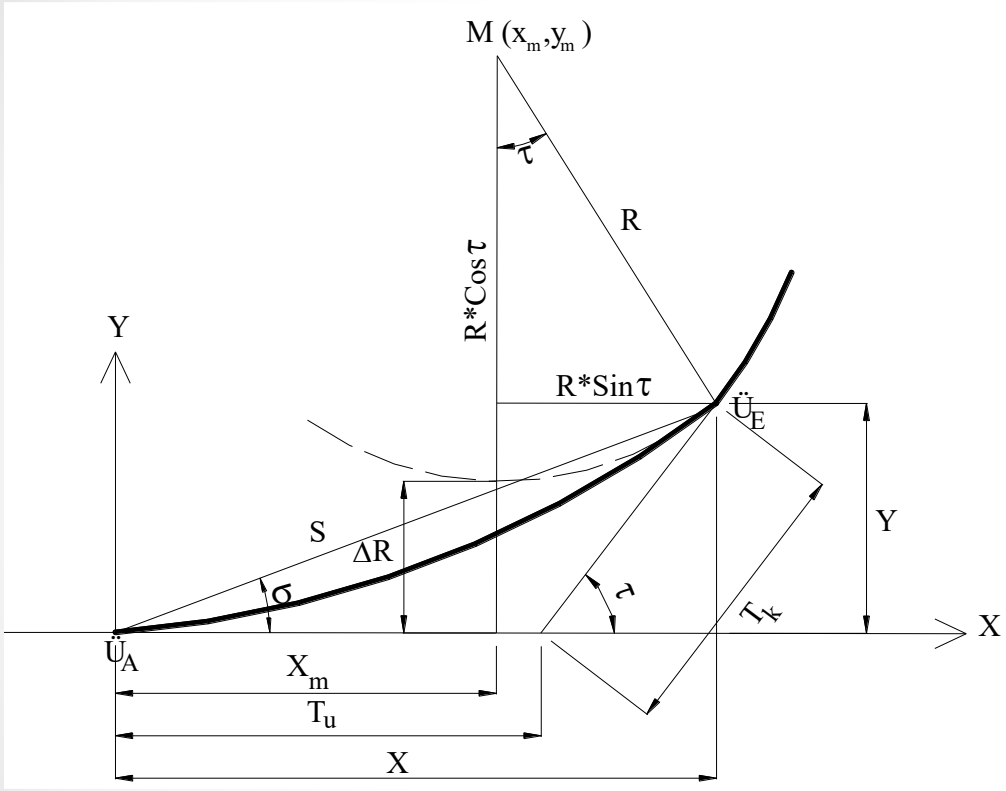


Clothoid : $A^2=L*R$

Lemniskat : $S=A^2*\sin(2*\sigma)$

Cubic Parabol : $$Y=\frac{X^3}{6*C}=\frac{X^3}{6*R*L}$$

TRANSITION CURVES



$$A^2 = L * R$$

$$\tau = \frac{L}{2 * R}$$

$$Y = \frac{L^3}{6 * A^2} - \frac{L^7}{336 * A^6} + \frac{L^{11}}{42240 * A^{10}} - \text{.....}$$

$$X_m = X - R * Sin\,\tau$$

$$Y_m = Y + R * Cos\,\tau$$

$$T_k = \frac{Y}{Sin\,\tau}$$

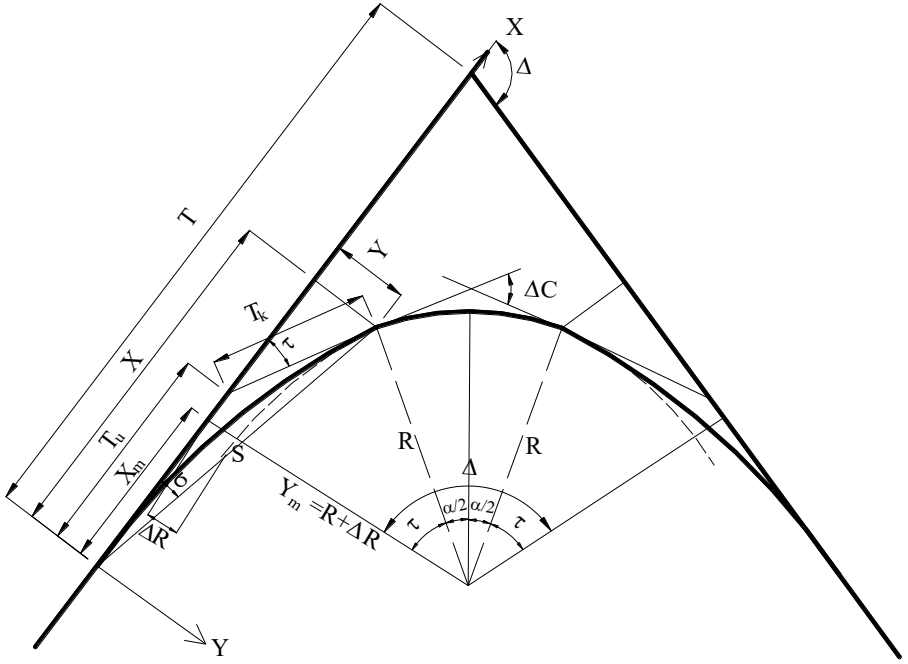
$$T_u = X - Y * Cot\,\tau$$

$$S = \sqrt{X^2 + Y^2}$$

$$\tan\sigma = \frac{Y}{X} \qquad \sigma = Arc\left(\frac{Y}{X}\right)$$

$$\Delta R = Y_m - R$$

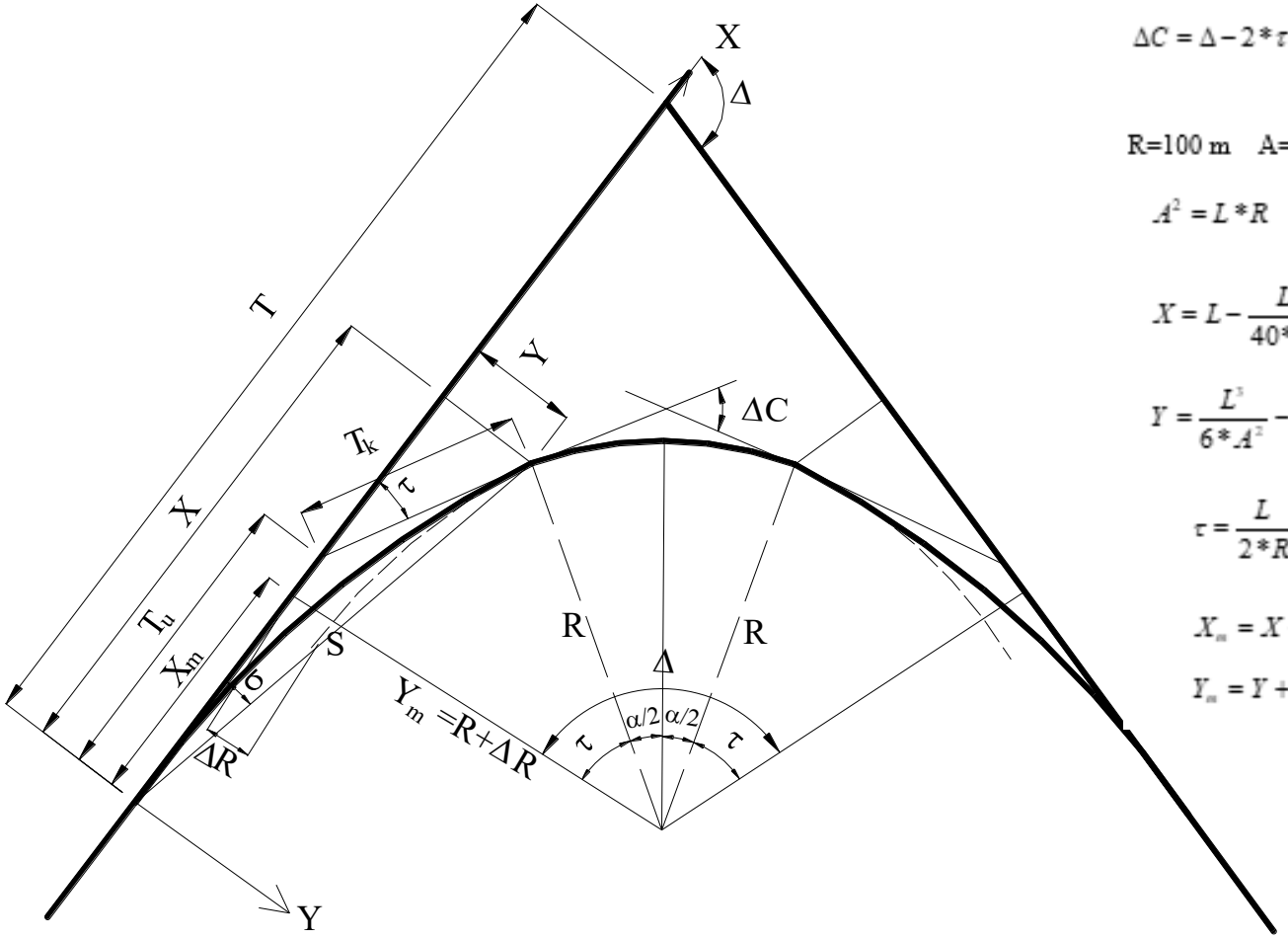
TRANSITION CURVES



The elements of the clothoid can be listed as:

- L : Length of the clothoid (also can be notationed as L_t as the transition curve length) (m),
- X : The abscissa of the end point of the clothoid (SC) with respect to its starting point (TS) (m),
- Y : The ordinate of the end point of the clothoid (SC) with respect to its starting point (TS) (m),
- τ : The tangent angle of the clothoid (radians),
- X_m : The abscissa of the horizontal curve center (m),
- Y_m : The ordinate of the horizontal curve center (m),
- T_u : The length of the longer tangent of the clothoid (m),
- T_k : The length of the shorter tangent of the clothoid or the distance between the tangent of the curve to the common tangent point SC (m),
- ΔR : Displacement of the horizontal curve (m),
- S : Clothoid chord (m),
- σ : Direction angle of the clothoid chord (degree),
- $\overline{TS}$: The tangent length of the clothoid or the distance between the start and the PI of the horizontal curve (m),
- ΔC : Divergence angle of the shifted curve (degree),
- Dc : Length of the shifted curve (m) ($Dc=CS_{km}-SC_{km}$).

TRANSITION CURVES



$\Delta C = \Delta - 2*\tau$ $T = t + X_m$ $t = (R+\Delta R)*\tan\left(\frac{\Delta}{2}\right)$

R=100 m A=70 Δ=32,77°

$A^2 = L*R$ L= 49 m

$X = L - \frac{L^5}{40*A^4} + \frac{L^9}{3456*A^8} - \ldots\ldots$ X= 48,71 m

$Y = \frac{L^3}{6*A^2} - \frac{L^7}{336*A^6} + \frac{L^{11}}{42240*A^{10}} - \ldots\ldots$ Y= 3,99 m

$S = \sqrt{X^2 + Y^2}$

$\tau = \frac{L}{2*R} = 14{,}04$ $\Delta R = Y_m - R = 1{,}00$

$X_m = X - R*Sin\tau = 24{,}45$ $\Delta C = 4{,}69°$

$Y_m = Y + R*Cos\tau = 101$ $D_c = 8{,}19m$

Transition curve length

$$L_t = \frac{V_p^3}{46{,}7 \cdot R \cdot p'}$$

TRANSITION CURVES

Selection of the clothoid flatness

Optical condition $\frac{R}{3} \leq A \leq R$

Dynamical condition $A_{min} = 0{,}19\sqrt{V_p^3}$

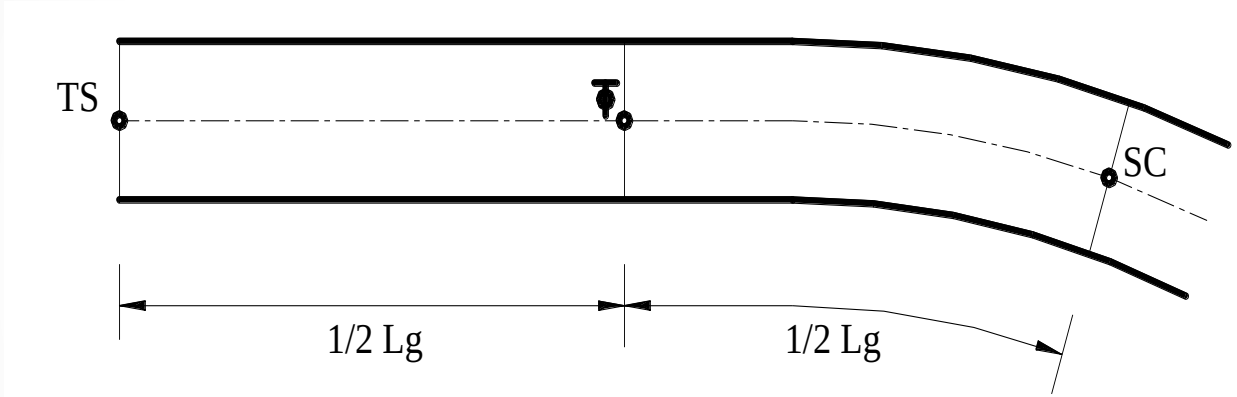
Superelevation condition $A_{min} = \sqrt{\frac{b}{2} \cdot \frac{(q_{min} + q_{max})}{0{,}005} \cdot R}$

Gometrical Condition $\frac{Lg}{2} \leq \frac{D}{2}$

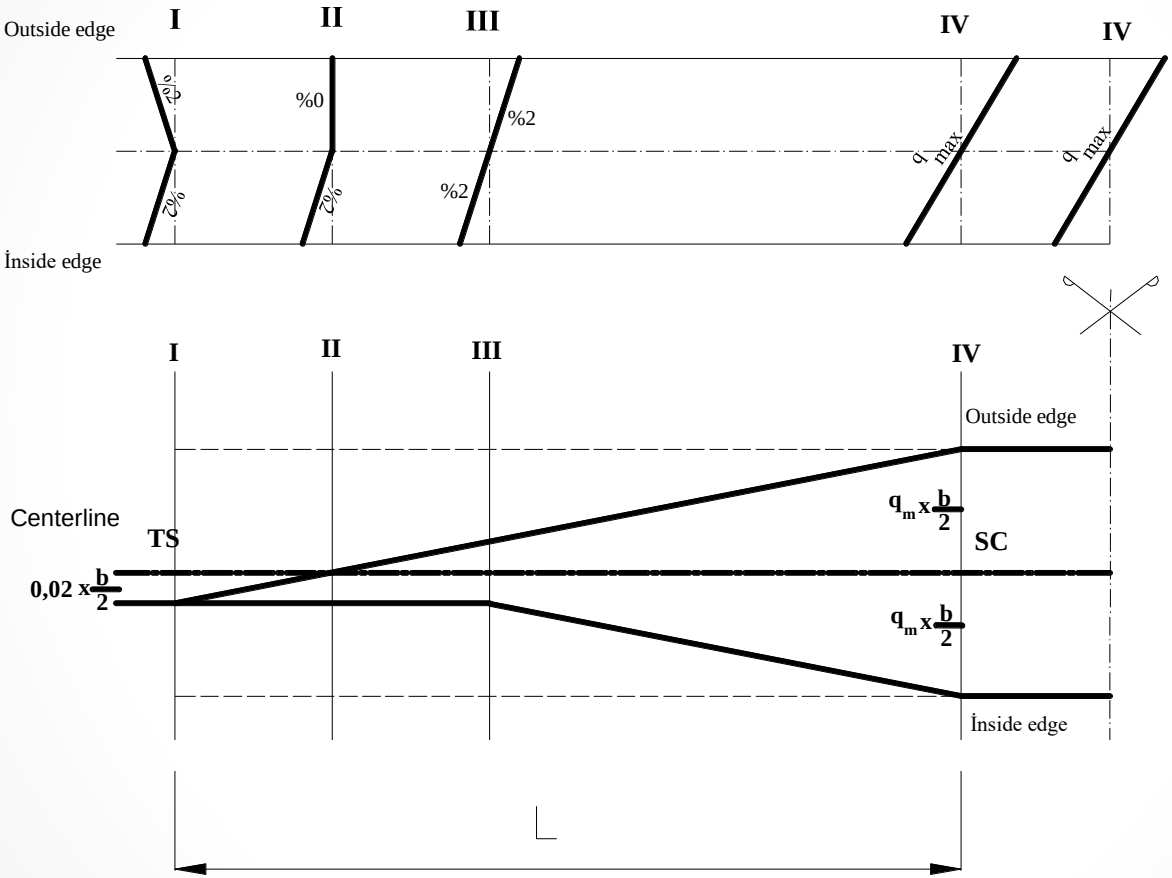
$Lg \geq L_{d,min} = 45m$

TRANSITION CURVES

TS point of clothoid falls on the point ½ *Lg before Φ, SC point falls on the point ½*Lg after Φ. On the symmetrical part of the curve, point CS falls on the point ½ *Lg before Ŧ, ST falls on the point ½ *Lg after Ŧ.

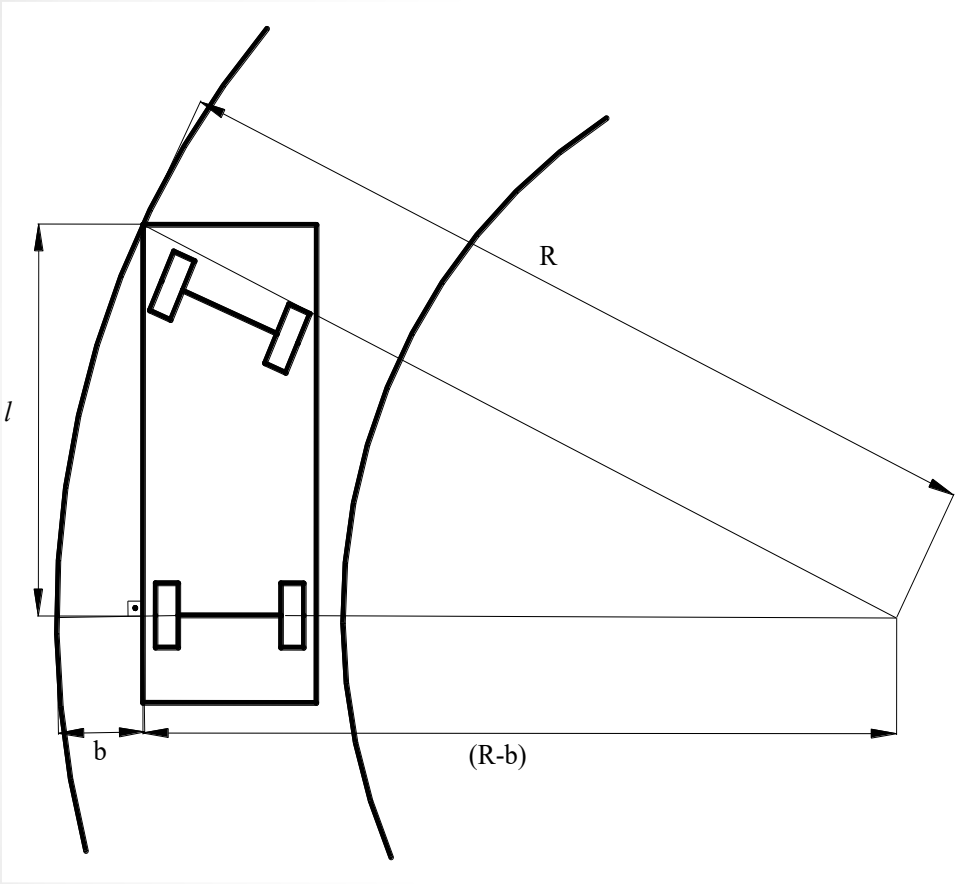


Implimentation of Superelevation with Transition Curve



TRANSITION CURVES

Widening of the Horizontal Curves



$$(R-b)^2 + l^2 = R^2$$

$$R^2 - 2*R*b + b^2 + l^2 = R^2$$

$$b^2 + l^2 = 2*R*b$$

b^2 is neglected

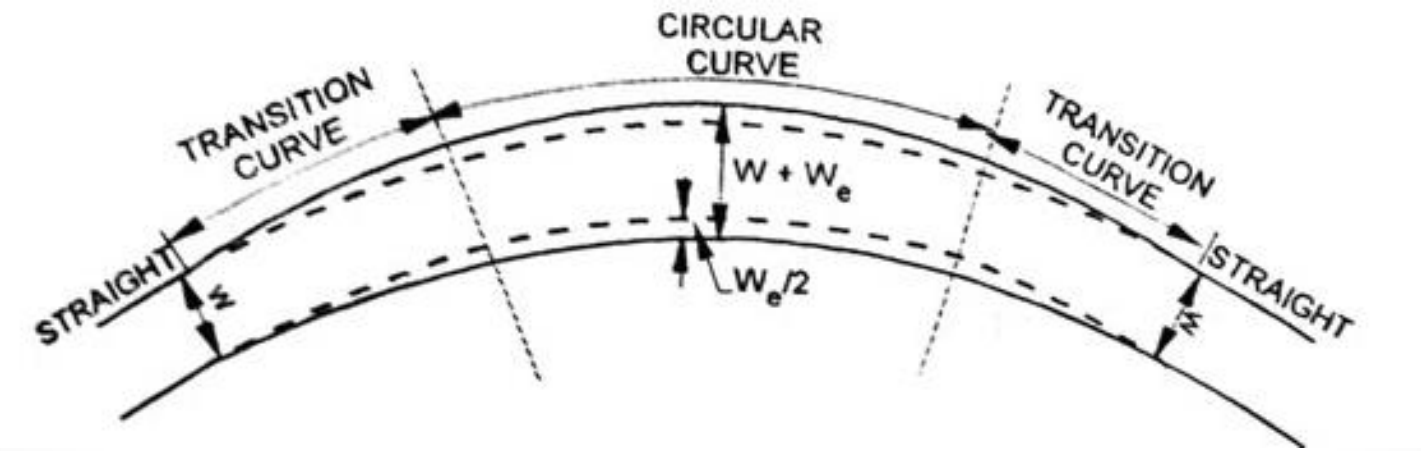
$$b = \frac{l^2}{2*R}$$

$$b = \frac{n*l^2(m)}{2*R(m)} + \frac{0{,}05*V_p(km/sa)}{\sqrt{R}}$$

Widening implimented if R < 200 m.

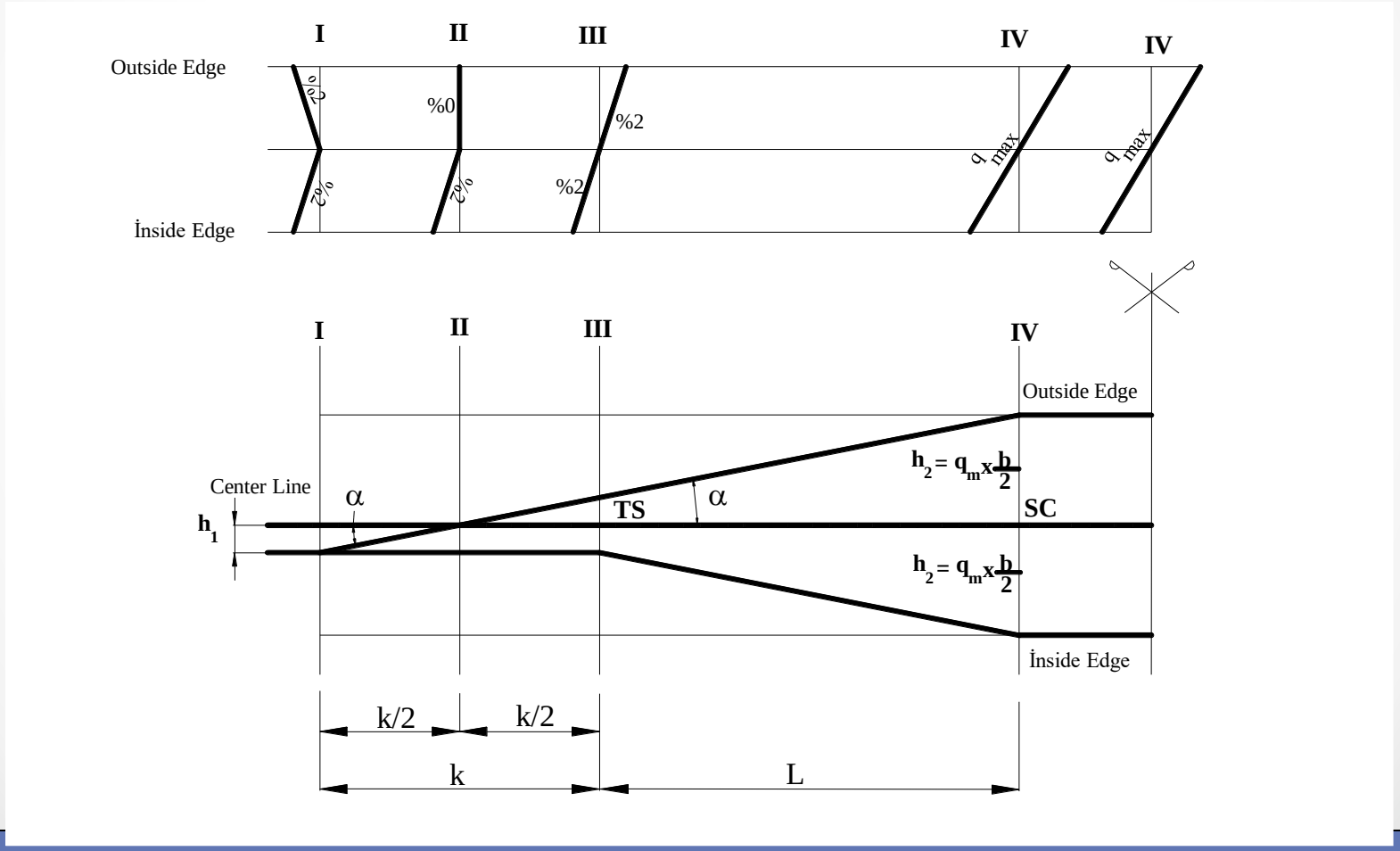
TRANSITION CURVES

Widening the curves

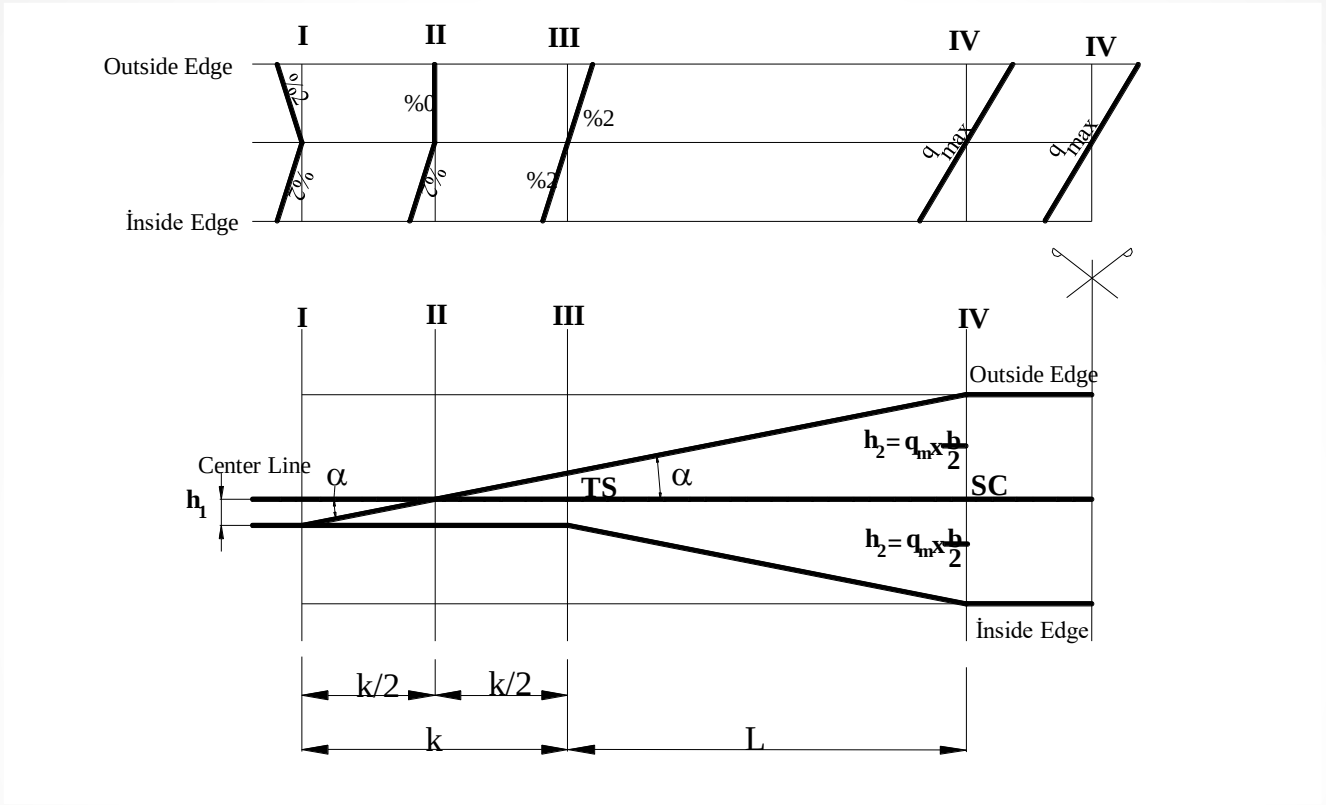


Calcultion of Ramp Length for Horizantal Curve

As the curvature starts to increase as entering the transition curve, the centrifugal force also starts to affect the vehicle. To counter this negative effects, the highway is desired to reach the single slope on this point



TRANSITION CURVES



$$\tan\alpha = \frac{h_2}{k/2 + L} = \frac{h_1}{h_2} \qquad \frac{h_1}{h_2} = \frac{k/2}{k/2 + L} \qquad h_1 * k/2 + h_1 * L = h_2 * k/2 \qquad h_1 * L = k/2 * (h_2 - h_1)$$

$$k = \frac{2 * h_1 * L}{h_2 - h_1}$$